

Toward an Independent Closure Derivation of the 255° Attractor in GEM

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Abstract

This note presents the clearest current formulation of the 255° attractor problem in General Emergence Mechanics (GEM). The effective phase free energy is decomposed into four parts: a Z_3 -symmetric orbifold term, a geometric chiral-bias term, a coherence-stiffness term, and a residual closure sector. Within this structure, a perturbative expansion around the baseline 240° minimum yields the conditional shift formula

$$\delta = \frac{\lambda(\theta_0 - 4\pi/3)}{9\lambda_3 - 2k + \lambda}.$$

The remaining open task is to derive the reference phase θ_0 and the stiffness λ independently from the full admissible free-energy landscape rather than inserting them phenomenologically. The note does not claim a completed first-principles proof of the 255° lock. Rather, it clarifies the effective structure of the problem and states precisely what the closure programme must still achieve for the attractor to become a true output of the theory.

Keywords: General Emergence Mechanics, GEM, closure sector, coherence stiffness, orbifold symmetry, chiral bias, 255° attractor, phase locking

1 Introduction

One of the central open questions in the development from Modal Theory to GEM is whether the phase attractor at 255° can be derived as a genuine output of the theory rather than inserted as a stabilising assumption. The purpose of this note is to state the current best version of that problem in a mathematically explicit and honest form.

The key issue is not whether a stiffness term *can* stabilise a minimum near 255°. That much is straightforward. The real issue is whether the preferred phase centre θ_0 and the coherence stiffness λ can both be obtained independently from the full admissible free-energy landscape. If they can, then the 255° lock becomes a derived feature. If they cannot, then the attractor remains a phenomenological closure ansatz.

This note therefore has a limited aim. It does not claim to complete the derivation. It clarifies the effective structure of the problem and identifies the independent closure derivation of θ_0 and λ as the true remaining task.

2 Effective Free Energy Decomposition

The coarse-grained phase free energy is written as

$$F_{\text{eff}}(\theta) = F_{Z_3}(\theta) + F_{\text{bias}}(\theta) + F_{\text{stiff}}(\theta) + F_{\text{closure}}(\theta). \quad (1)$$

The individual terms are:

1. Z_3 -symmetric orbifold term

$$F_{Z_3}(\theta) = -\lambda_3 \cos(3\theta),$$

which provides the baseline threefold minima at 0° , 120° , 240° .

2. Geometric chiral-bias term

$$F_{\text{bias}}(\theta) = -k(\cos \theta + \sqrt{3} \sin \theta).$$

3. Coherence-stiffness term

$$F_{\text{stiff}}(\theta) = \frac{\lambda}{2}(\theta - \theta_0)^2,$$

where $\lambda > 0$ is the effective phase rigidity and θ_0 is the closure-selected reference phase.

4. Closure sector

$$F_{\text{closure}}(\theta),$$

which encodes residual fluctuation effects, orbifold mode-sum corrections, and vacuum-amplitude contributions.

Equation (1) is the essential structural statement. The 255° problem concerns the full admissible balance of all four terms.

3 Coherence Stiffness

The stiffness λ arises from integrating out fast fluctuations around the locked vacuum. Schematically,

$$\lambda = \lambda(\rho_0, \Lambda_{\text{orb}}, g_{\text{mode}}, \dots) \approx \rho_0^2 \Lambda_{\text{orb}}^2 \cdot \mathcal{G}(g_{\text{mode}}).$$

Physically, λ represents the shell's resistance to phase distortion — the effective elasticity of the phase-locked vacuum.

4 Perturbative Shift from the Baseline Minimum

We expand around the baseline Z_3 minimum at $\theta = 4\pi/3$:

$$\theta = \frac{4\pi}{3} + \delta, \quad |\delta| \ll 1.$$

To first order the stationarity condition yields

$$(9\lambda_3 - 2k + \lambda)\delta + \lambda \left(\frac{4\pi}{3} - \theta_0 \right) = 0.$$

Solving for the shift gives

$$\boxed{\delta = \frac{\lambda(\theta_0 - 4\pi/3)}{9\lambda_3 - 2k + \lambda}.$$

This is the central perturbative result. It shows explicitly how a displaced closure-selected phase centre θ_0 and finite stiffness λ would shift the baseline minimum.

5 The Remaining Closure Task

The real unfinished task is to derive θ_0 and λ independently from the full admissible free-energy landscape (vacuum amplitude ρ_0 , orbifold mode sum, and fluctuation averaging).

The working closure hypothesis is that, in the physically relevant regime, this procedure selects

$$\theta_0 = \frac{4\pi}{3} + \frac{\pi}{12} = 255^\circ.$$

If confirmed by explicit closure, the 15° shift is not an arbitrary input but the natural outcome of the geometric-dynamical machinery.

6 Conceptual Upgrade

The main conceptual advance is this:

θ_0 and λ are not free parameters to be chosen.

They are coordinates of the deepest admissible coherence basin.

The 255° lock, if ultimately established, should emerge because it is stationary, locally stable, and compatible with the vacuum amplitude and orbifold structure — not because it was inserted by hand.

7 What Has and Has Not Been Shown

What *has* been shown: - The correct effective free-energy structure of the closure problem, - The perturbative mechanism by which a displaced centre shifts the baseline minimum, - The precise role played by stiffness and chiral bias.

What has *not* yet been shown: - A fully explicit analytic derivation of λ , - A fully explicit analytic derivation of $\theta_0 = 255^\circ$, - A complete closed-form expression for F_{closure} .

This note therefore does not claim a final proof. It clarifies the best current machinery and states exactly what remains to be done.

8 Conclusion

The 255° attractor problem can now be stated more cleanly. The effective free energy is naturally decomposed into Z_3 -symmetric, chiral-bias, coherence-stiffness, and closure sectors. The perturbative shift formula shows how a closure-selected reference phase would displace the baseline 240° minimum.

The key unresolved issue is no longer hidden: θ_0 and λ must still be derived independently from the full admissible free-energy landscape. Until that is done, the 255° lock remains a strong candidate output of closure rather than a finished first-principles theorem.

This formulation turns the 255° question into a clear research programme rather than an unresolved assertion.